Cyclic Photophosphorylation

- It occurs in the stroma lamellae of chloroplast because here PS II as well as NADP reductase enzyme are absent.
- When PS I (P_{700}) is activated by light, it is raised to high energy level and emits a pair of electrons.
- These electrons are captured by the primary acceptor, which passes them on to Ferredoxin, Plastoquinone, cytochrome complex and plastocyanin.
- Finally, the de-energised electrons come back to P_{700}.
- Since the electrons come back to P_{700}, this process of ATP formation is known as cyclic photophosphorylation.

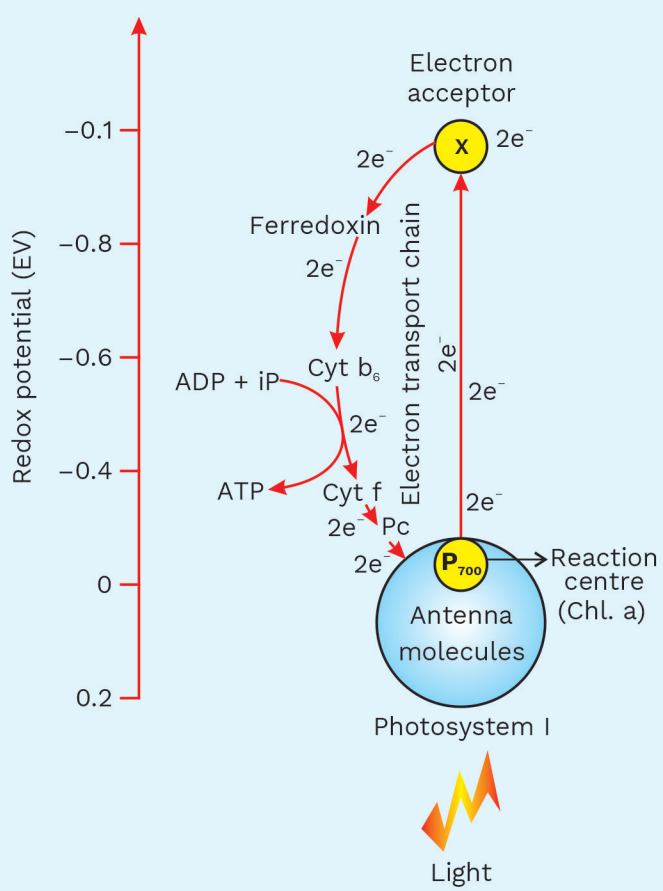

FIG. CYCLIC PHOTOPHOSPHORYLATION

Previous Year's Question

Energy is transformed from the light reaction step to the dark reaction step by
(1) Chlorophyll
(2) ADP
(3) ATP
(4) RuBP

Definition

Photocentres or reaction centres: The chlorophyll a molecules which expel electrons after excitation to electron acceptors for the generation of ATP in PS I and PS II.

Previous Year's Question

In both respiration and photosynthesis ATP synthesis is an oxidative process that uses energy of
(1) Electrons
(2) Ferredoxin
(3) Cytochromes
(4) Carbon dioxide

Definition

Non-cyclic photophosphorylation: Electrons move in a closed circle. Here electrons expelled from P_{700} (reaction centre) return to it.